

Pulmonary vein isolation in atrial fibrillation patients guided by a novel local impedance algorithm: 1-year outcome from the CHARISMA study

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Abstract

Background: Highly localized impedance (LI) measurements during atrial fibrillation (AF) ablation have recently emerged as a viable real-time indicator of tissue characteristics and durability of the lesions created. We report the outcomes of acute and long-term clinical evaluation of the new DirectSense algorithm in AF ablation. **Methods:** Consecutive patients undergoing AF ablation were included in the CHARISMA registry. RF delivery was guided by the DirectSense algorithm, which records the magnitude and time-course of the impedance drop. The ablation end-point was pulmonary vein isolation (PVI), as assessed by the entrance and exit block. **Results:** 3556 point-by-point first-pass RF applications of >10 s duration were analyzed in 153 patients (mean age = 59 ± 10 years, 70% men, 61% paroxysmal AF, 39% persistent AF). The mean baseline LI was 105 ± 15 Ω before ablation and 92 ± 12 Ω after ablation ($p < .0001$). Both absolute drops in LI and the time to LI drop (LI drop/τ) were greater at successful ablation sites ($n = 3122$, 88%) than at ineffective ablation sites ($n = 434$, 12%) ($14 \pm 8 \Omega$ vs $6 \pm 4 \Omega$, $p < .0001$ for LI; $0.73 [0.41-1.25] \Omega/s$ vs $0.35 [0.22-0.59] \Omega/s$, $p < .0001$ for LI drop/τ). No major complications occurred during or after the procedures. All PVs had been successfully isolated. During a mean follow-up of 366 ± 130 days, 18 patients (11.8%) suffered an AF/atrial tachycardia recurrence after the 90-day blanking period.

Abbreviations: ADT, antiarrhythmic drug therapy; AF, atrial fibrillation; AFL, atrial flutter; AI, ablation index; AT, atrial tachycardia; CF, contact force; CI, confidence interval; FU, follow-up; HR, hazard ratio; ILD, inter-lesion distance; LI, local impedance; PV, pulmonary vein; PVI, pulmonary vein isolation; RF, radiofrequency; ROC, receiver operating characteristic; VT, ventricular tachycardia.

Clinical Trial Registration

Catheter Ablation of Arrhythmias with a High-Density Mapping System in Real-World Practice (CHARISMA). URL: <http://clinicaltrials.gov/Identifier: NCT03793998>

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Conclusion: The magnitude and time-course of the LI drop during RF delivery were associated with effective lesion formation. This ablation strategy for PVI guided by LI technology proved safe and effective and resulted in a very low rate of AF recurrence over 1-year follow-up.

KEYWORDS

atrial fibrillation, catheter ablation, DirectSense, lesion formation, local impedance

1 | INTRODUCTION

Catheter ablation through pulmonary vein (PV) isolation is an established effective treatment for both paroxysmal and persistent atrial fibrillation (AF) and is recommended during all AF catheter-ablation procedures.^{1,2} Currently, one of the prevailing approaches to the electrical isolation of PVs is point-by-point radiofrequency (RF) ablation; this is achieved by creating linear lesions encircling the PVs and is guided by conventional circular mapping catheters or a small basket array and nonfluoroscopic mapping systems.³ Complete electrical isolation of the PVs, with documented entrance and/or exit block, has better outcomes on rhythm than incomplete isolation.⁴ However, long-term atrial arrhythmia-free survival is still unsatisfactory.⁵ In addition, the high rate of PV reconnection after ablation seems to contribute to ablation failures.^{6,7} Many advances in catheter technology have recently shown the potential to transform the approach to arrhythmia ablation⁸ and to assess its completeness in terms of transmural and durability of the sets of lesions created.⁴ These include systems designed to assess the real-time effectiveness of ablation,⁹ such as DirectSense, a novel technology that measures local impedance (LI) through mini-electrodes on the tip of a dedicated ablation catheter (IntellaNAV Mifi OI, Boston Scientific). The use of highly localized impedance measurements to provide insight into tissue characteristics and their real-time evaluation seem to be helpful to precisely assess the electrical contact of the catheter and tip stability.^{10,11} However, although preliminary data on acute success and ablation endpoints have been encouraging, clinical evidence on the long-term outcome in a large population of AF patients is still lacking. Here, we report the outcomes of our acute and long-term clinical evaluation of an ablation strategy guided by the magnitude and time-course of the impedance drop, as measured through the DirectSense algorithm, during RF delivery in consecutive patients undergoing AF ablation. The characteristics of local impedance at different ablation spots were also evaluated.

2 | METHODS

2.1 | Patient population and study design

The Catheter Ablation of Arrhythmias with High-Density Mapping System in the Real World Practice (CHARISMA) ([ClinicalTrials.gov](https://clinicaltrials.gov)

Identifier: NCT03793998) is a prospective, single-arm, multicenter cohort study designed to describe clinical practice regarding the approach to the ablation of different arrhythmias. The study complied with the Declaration of Helsinki, the locally appointed ethics committee approved the research protocol, and informed consent was obtained from all patients. In this paper, we present the analysis of 153 consecutive patients indicated for AF ablation who underwent high-resolution mapping and ablation guided by a local impedance algorithm in eight Italian centers. All patients were followed up at the same hospital, from the time of the ablation to the last follow-up visit.

2.2 | Ablation procedure

After completion of the baseline evaluation, patients underwent ablation in accordance with standard clinical practice guidelines.¹ All procedures were performed under conscious sedation or general anesthesia. Vitamin K antagonist treatment was not interrupted, while non-Vitamin K anticoagulants were omitted on the morning of the procedure. A decapolar catheter (Dynamic XT, Boston Scientific) was used to cannulate the coronary sinus. After single or double transseptal punctures under fluoroscopic guidance, intravenous unfractionated heparin boluses were administered, to maintain an activated clotting time of >300 s, and then the basket mapping catheter (Orion, Boston Scientific) and the ablation catheter (IntellaNav MiFi OI, Boston Scientific) were inserted through steerable sheaths. The Orion catheter was used in combination with the Rhythmia HDx mapping system (Rhythmia, Boston Scientific) to create a three-dimensional electroanatomical voltage and activation map of the left atrium. Mapping and ablation were carried out in sinus rhythm; in patients in AF, electrical cardioversion was performed to restore sinus rhythm. The ablation target for all patients was PVI only. PVI was performed with radiofrequency energy in a point-by-point ablation pattern by the open-irrigated ablation catheter. Point-by-point RF delivery was performed in such a way as to create contiguous ablation spots encircling the PVs. In patients undergoing de novo AF ablation, contiguous ablation spots were created around the PV Ostia where the local electrograms did not show near-field PV signals. In redo procedures, the ablation spots were deployed where conduction gaps were found after evaluation of the initial voltage/activation map. In cases of atrial tachycardia, the ablation spots were

deployed where critical points were found after evaluation of the initial voltage/activation map and the execution of entrainment maneuvers, when appropriate. Ablation on the intervenous carinae was not performed as part of the initial lesion set unless it was needed to achieve PV isolation. Ablation was guided by the magnitude and time-course of impedance drop during RF delivery. The recommended maximum distance between each ablation spot (center-to-center) was ≤ 6 mm. RF applications were targeted to a minimum LI drop of 10Ω within 30 s and were stopped when a maximum cutoff LI drop of $\geq 40\Omega$ was observed.

Successful single RF ablation was defined as a reduction in signal voltage by at least 50% and the inability to capture local tissue on pacing.^{10,12} Targeted signal was evaluated through the EGM recordings from the micro-electrodes (i.e., the largest MiFi signal). Pacing maneuvers were performed from each lesion site at the end of the PVs circle. Pacing output was set at 10 V at 2 ms, with a pacing cycle length of 600 ms. Unsuccessful lesions were defined as those which resulted in continued ability to capture tissue.^{10,12} Power settings were at the individual operator's discretion, within the range of 30 to 35 W, depending on the left atrial segment considered. The endpoint of ablation was PVI, as assessed on the basis of entry and exit block by means of the 64-pole Orion catheter placed sequentially in each of the PVs. In the absence of first-pass PV isolation (i.e., no isolation upon completion of the encirclement of ipsilateral veins), PV isolation was accomplished by means of additional RF applications at the investigator's discretion.

2.3 | Local impedance measurement

Baseline tissue impedance and impedance drop for each ablation lesion were measured both from the micro-electrodes of the catheter and from the Maestro generator (Maestro 4000, Boston Scientific) (generator impedance [GI]). The user interface displays the maximum LI value of all three miniature-electrode LI measurements. The characteristics of measurement functioning have already been described elsewhere.¹⁰

When available, videos of the ablation procedures were exported from the mapping system, displaying the procedure in real-time. RF current applications were then retrospectively analyzed. For the purpose of this analysis, we excluded all lesions with the following characteristics: catheter instability, RF deliveries outside PVs, interference with the Orion catheter, and RF application lasting less than 10 s.¹²

2.4 | Follow-up

Complications were reported on the case report form and collected during follow-up. After ablation, anticoagulation and anti-arrhythmic drug therapy (ADT) were continued. At 3 months, anticoagulation was continued according to stroke risk, whereas ADT was continued at the discretion of the treating physician.

Clinical evaluation and ECG were performed at 1, 3, 6, and 12 months. Holter ECG was performed at 3, 6, and 12 months post-ablation or in the case of symptoms. Arrhythmia recurrences within the first 3 months (blanking period) were classified as early recurrences and were not considered procedural failures. Freedom from recurrence was defined as the 1-year absence of AF, atrial tachycardia, and atrial flutter (AF/AT/AFL) lasting continuously for >30 s, as recorded by any postablation electrocardiogram (ECG) modality.¹³

2.5 | Statistical analysis

Descriptive statistics are reported as means $\pm$ SD for normally distributed continuous variables, or medians with 25th to 75th percentiles in the case of skewed distribution. Normality of distribution was tested by means of the non-parametric Kolmogorov-Smirnov test. Differences between mean data were compared by means of a *t* test for Gaussian variables, and the *F* test was used to check the hypothesis of equality of variance. The Mann-Whitney nonparametric test was used to compare non-Gaussian variables. Differences in proportions were compared by applying χ^2 analysis or Fisher's exact test, as appropriate.

The Kaplan-Meier method was used to analyze estimates of time to recurrences during follow-up in patients stratified by AF type and AF procedure; differences between groups were analyzed by means of the log-rank test. Hazard ratios (HRs) and their 95% confidence intervals (CIs) were computed by means of Cox regression models, in which baseline parameters were considered as fixed covariates and endpoint events were considered as time-dependent covariates. The proportional hazard assumption was assessed by means of Schoenfeld residuals. A *p* value $<.05$ was considered significant for all tests. All statistical analyses were performed by means of STATISTICA software, version 7.1 (StatSoft, Inc.).

3 | RESULTS

3.1 | Study population

One-hundred fifty-three consecutive patients were enrolled: patients had an indication mostly for paroxysmal AF ablation ($n = 94$, 61.4%) and the majority of procedures were de novo PVI ($n = 96$, 62.7%). In about half of the patients ($n = 73$, 47.7%) the first occurrence of atrial arrhythmia was reported more than 1 year before the procedure. Besides AF, 21.6% of patients had a history of atrial flutter (AFL), and 5.2% a history of AT. The mean procedure duration was 136 ± 47 min and the mean fluoroscopy time was 17.1 ± 8 min. A total of 7598 RF applications were delivered, with a median number of 44 27–66 ablation spots and a median RF pulse duration time of 33 24–48 s, without any steam popping. Baseline clinical characteristics are shown in Table 1.

3.2 | Outcome

At the end of the procedures, all PVs had been successfully isolated in all study patients. No major complications, including atrioesophageal fistula or tamponade, were reported during or after the procedures. Minor complications were reported in nine patients (5.9%) after the procedure: vascular complications at the puncture site in 5 (three groin hematomas, one pseudoaneurysm, and one arteriovenous fistula) and pericarditis with mild pericardial effusion in 4. Conservative treatment and medical therapy were effective in all cases. During a mean follow-up of 366 ± 130 days, 18 patients (11.8%) suffered an AF/AT recurrence after the 90-day blanking period. Specifically, 10 (6.5%) patients had AF recurrence only, 7 (4.6%) had AT recurrence only and 1 (0.7%) experienced both events. The recurrence rate was higher in patients with persistent AF (11 out of 59, 18.6%) than in those with paroxysmal AF (7 out of 94, 7.4%; $p = .0426$), whereas no differences emerged between procedure types (10 out of 96–10.4%—for de novo procedures vs. 8 out of 57–14%—for redo procedures, $p = .6052$). De novo paroxysmal AF procedures resulted in the lowest recurrence rate (4 out of 58, 6.9%), whereas redo persistent AF procedures showed the highest rate (5 out of 21, 23.8%).

Figure 1 reports the time to recurrence after the blanking period over 12-month follow-up (FU). The time to recurrence was slightly longer in paroxysmal AF (log-rank $p = .0474$ for persistent vs. paroxysmal AF, Figure 2A) and comparable between procedure types (log-rank $p = .5653$ for redo vs. de novo procedures, Figure 2B).

3.3 | Characteristics of local tissue impedance

In 105 (69%) cases with complete video recording, we analyzed a total of 3556 point-by-point first-pass RF applications of >10 s duration delivered around PVs and with high-quality data. The mean baseline LI was $105 \pm 15 \Omega$ vs. $90.1 \pm 11 \Omega$ for GI ($p < .0001$) before ablation and $92 \pm 12 \Omega$ for LI vs. $85.9 \pm 9 \Omega$ for GI ($p < .0001$) after ablation ($p < .0001$ for both LI and GI before ablation vs. after ablation, mean absolute LI drop of $13 \pm 8 \Omega$ vs. $4.2 \pm 5 \Omega$ for GI, $p < .0001$; mean percentage LI drop of $12 \pm 7\%$ vs. $4.4 \pm 5\%$ for GI, $p < .0001$) Figure 3A,B; the mean RF delivery time was 31 ± 23 s. Successful ablation sites were predicted by both baseline LI and LI drop (OR = 1.03, 95% CI: 1.02–1.04, $p < .0001$ for baseline LI; OR = 1.26, 95% CI: 1.22–1.29, $p < .0001$ for LI drop). Absolute drops in impedance were larger at successful ablation sites ($n = 3122$, 88%) than at ineffective ablation sites ($n = 434$, 12%), and through LI than through conventional GI ($14 \pm 8 \Omega$ vs. $6 \pm 4 \Omega$, $p < .0001$ for LI; $4.3 \pm 5 \Omega$ vs. $3.1 \pm 5 \Omega$, $p < .0001$ for GI) Figure 3C. Area under the receiving operating curve (AUC) in predicting successful ablation sites was larger for LI drop than GI drop (AUC = 0.806, standard error = .00774, 95% CI: 0.793–0.819 for LI drop vs. AUC = 0.582, standard error = .0193, 95% CI: 0.557–0.606 for GI drop; difference = 0.224, standard error = .0208, $p < .0001$). Every 5-point increment in LI drop was associated with successful ablation, with an OR of 3.13 (95% CI: 2.7–3.6,

TABLE 1 Baseline characteristics and procedural parameters

Parameter	n = 153
Age, years	59 ± 10
Male sex, n (%)	107 (69.9)
Indication for ablation	
• Paroxysmal AF, n (%)	• 94 (61.4)
• Persistent AF, n (%)	• 59 (38.6)
Ablation strategy	
• De novo PVI, n (%)	• 96 (62.7)
• Redo PVI, n (%)	• 57 (37.3)
Time to first arrhythmic episode greater than 1 year, n (%)	73 (47.7)
History of atrial flutter/atrial tachycardia, n (%)	41 (26.8)
LVEF, %	57 ± 7
LAD, mm	42 ± 7
Cardiomyopathy, n (%)	26 (17)
Hypertension, n (%)	69 (45.1)
Coronary artery disease, n (%)	5 (3.3)
BMI, kg/m ²	27.4 ± 5
History of heart failure, n (%)	8 (5.2)
COPD, n (%)	9 (5.9)
CKD, n (%)	4 (2.6)
ACE-ARB, n (%)	38 (24.8)
Beta-blockers, n (%)	64 (41.8)
Statin, n (%)	36 (23.5)
Diuretics, n (%)	21 (13.7)
Antiarrhythmics, n (%)	87 (56.9)
Procedure duration, min	136 ± 47
Fluoroscopy time, min	17.1 ± 8
RFC applications, n	50 ± 31
RFC duration time, s	31 ± 23
Mean power, W	35 ± 4
Complications during the procedure, n (%)	0 (0)
Minor complications	9 (5.9)
• Vascular complications	• 5
• Pericarditis	• 4

AF, atrial fibrillation; PVI, pulmonary vein isolation; LVEF, left ventricular ejection fraction; LAD, left atrial diameter; BMI, body mass index; RFC, radiofrequency catheter; COPD, chronic obstructive pulmonary disease; CKD, chronic kidney disease; ACE, angiotensin-converting enzyme; ARB, angiotensin receptor blocker.

$p < .0001$), reaching the highest point when a value larger than 15Ω was observed (Figure 4A). Baseline LI values were higher in paroxysmal AF than in persistent AF patients ($105.7 \pm 14 \Omega$ vs. $103.5 \pm 16 \Omega$, $p < .0001$). However, on considering de novo patients

FIGURE 1 Kaplan–Meier estimates of time to AT/AF recurrence after the blanking period over 12-month FU in the whole population. AF, atrial fibrillation; AT, atrial tachycardia; FU, follow-up

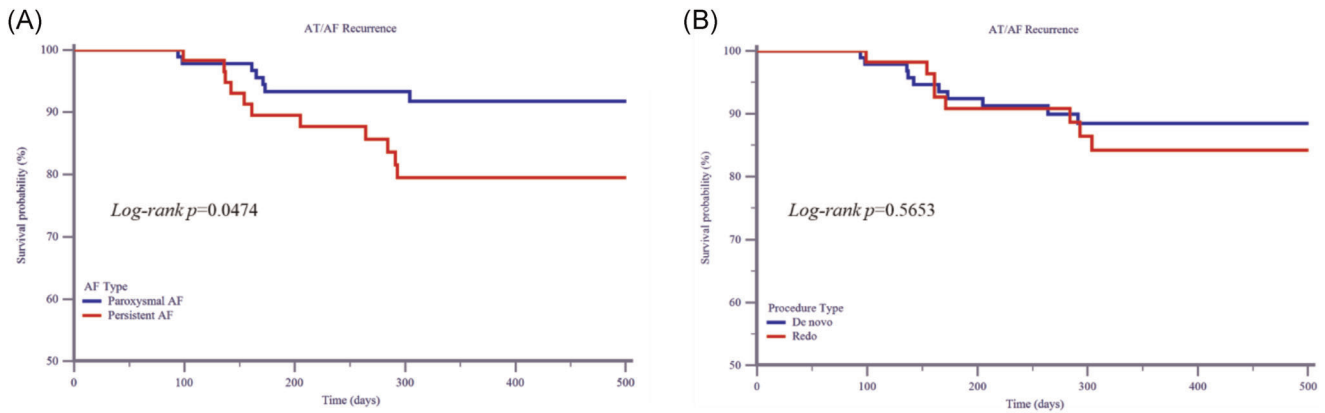
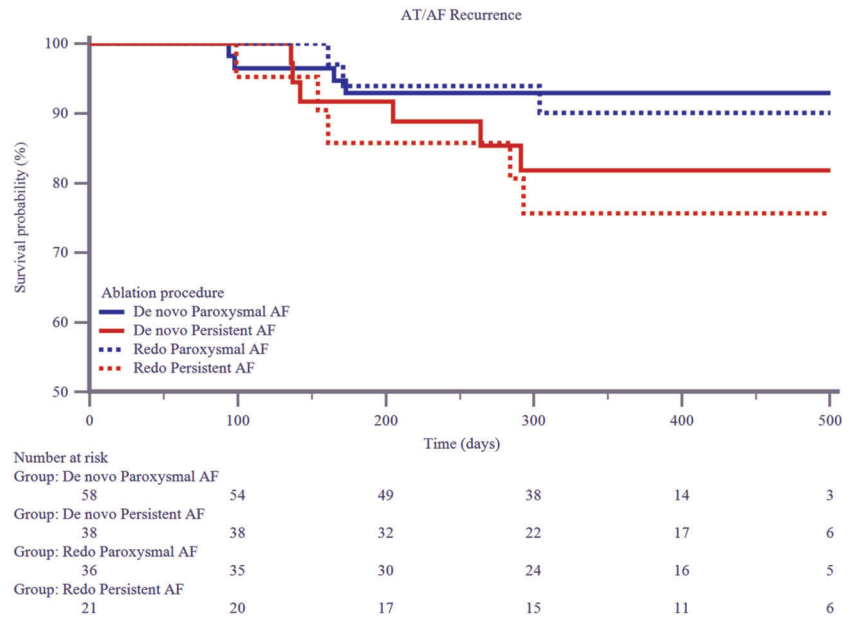


FIGURE 2 (A, B) Kaplan–Meier estimates of time to AT/AF recurrence after the blanking period over 12-month FU. (A) cases of paroxysmal and persistent AF. (B) de novo and redo procedures. AF, atrial fibrillation; AT, atrial tachycardia; FU, follow-up

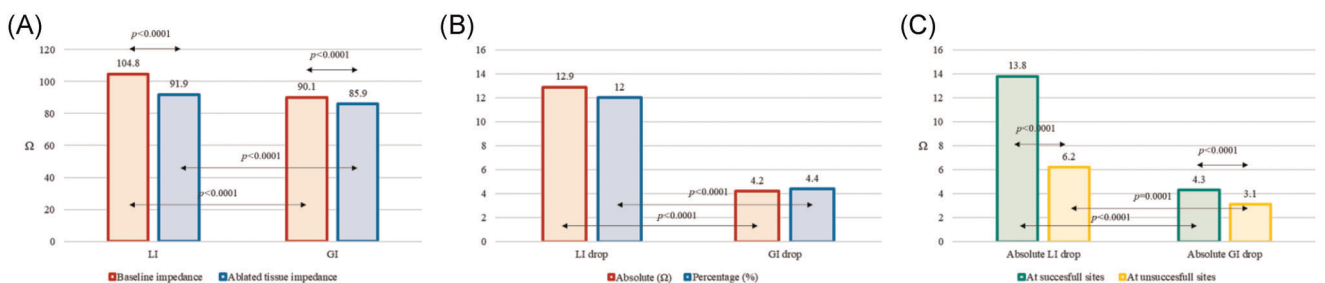


FIGURE 3 (A) Relationship between LI and GI before and after RF delivery. (B) Relationship between absolute and percentage values of LI drop and GI drop. (C) Absolute LI drop and GI drop according to successful and unsuccessful ablation sites. LI, localized impedance

only, no differences in baseline LI were found between these groups ($105.1 \pm 14 \Omega$ for paroxysmal AF vs. $105.2 \pm 16 \Omega$ for persistent AF, $p = .8301$), whereas the LI drop was lower in paroxysmal AF patients ($12.4 \pm 8 \Omega$ for paroxysmal AF vs. $13.4 \pm 8 \Omega$ for persistent AF, $p = .001$). The atrial substrate voltage was analyzed at 3014 (85%)

sites at which complete high-quality data were available for evaluation. Ablation spots were more frequently deployed in high-voltage areas (>0.5 mV; $n = 1739$, 57.4%) than in areas of intermediate voltage (0.1 – 0.5 mV; $n = 687$, 22.8%) or low voltage (<0.1 mV; $n = 597$, 19.8%). LI characteristics according to with

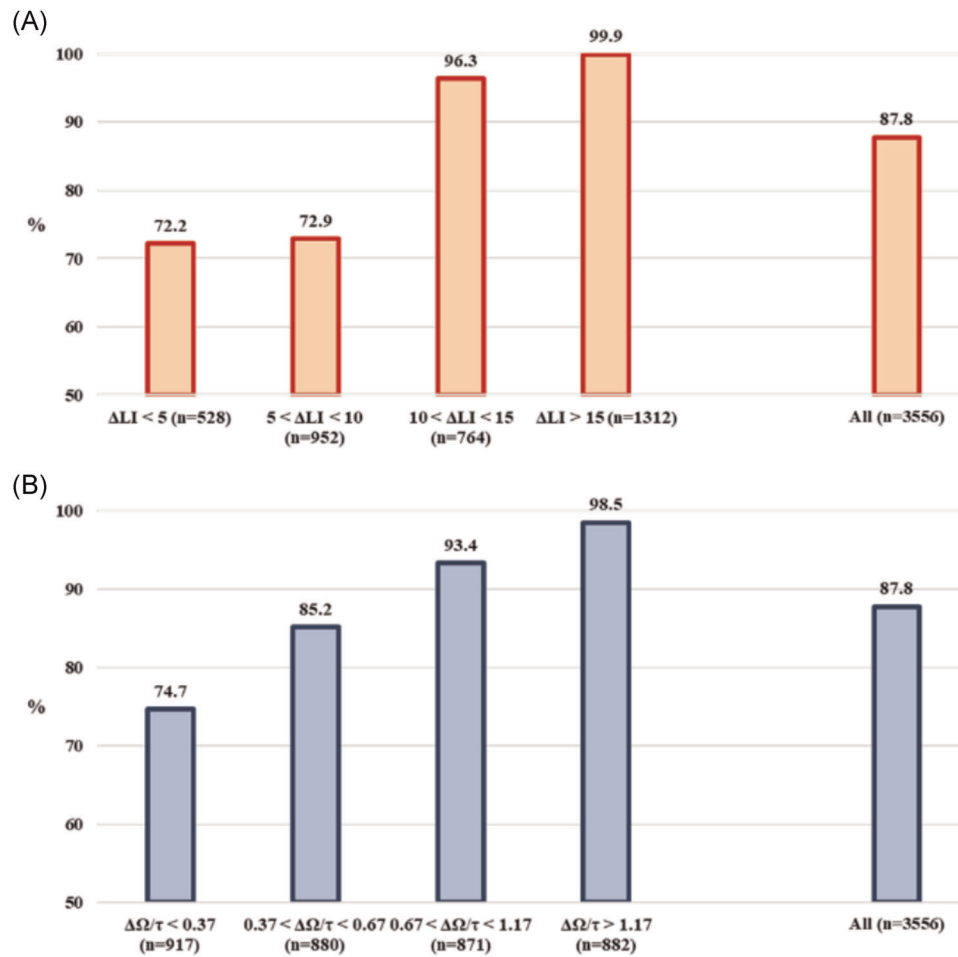


FIGURE 4 (A) Percentage of successful ablation spots according to LI drop values. (B) Percentage of successful ablation spots according to LI drop/τ values. LI, localized impedance

different voltage areas are shown in Figure 5. The mean time to drop (τ, defined as the time to LI drop at each ablation spot) was 18.7 ± 13 s, resulting in $71 \pm 35\%$ of the total ablation time per RF delivery. The median ratio between LI drop and τ was 0.67 [0.37–1.17] Ω/s. The median LI drop/τ values were higher at successful ablation sites than at unsuccessful ablation sites (0.73

[0.41–1.25] Ω/s at successful sites vs. 0.35 [0.22–0.59] Ω/s at unsuccessful sites, $p < .0001$) and a significant trend was observed from lower to higher LI drop/τ values (Figure 4B). On the basis of the ROC analysis, the cut-off value of LI drop/τ that best predicted successful RF delivery was 0.65 Ω/s (sensitivity 56.4%, specificity 81.8%, area under the curve 0.749, $p < .0001$) and a LI drop/τ greater than this

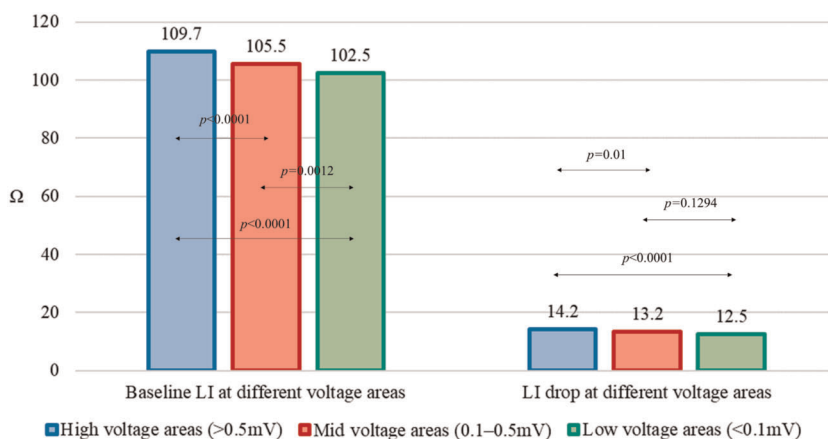


FIGURE 5 Baseline LI and LI drop according with different voltage areas

value was significantly associated with successful RF delivery, with an OR of 5.54 (95% CI: 4.31–7.11, $p < .0001$).

4 | DISCUSSION

4.1 | Main findings

In this single-arm prospective study, we performed catheter ablation of AF by means of a novel ablation catheter with local impedance-sensing capability. The ablation strategy, which was guided by local impedance information, had a 100% acute procedural effectiveness rate, without any steam pops or major complications, including atrioesophageal fistula or tamponade. The magnitude and time-course of the LI drop were strongly associated with effective ablation sites. In our mixed population of paroxysmal and persistent AF patients, the AT/AF recurrence-free rate without a blanking period was >88% at 12 months.

4.2 | Comparison with previous findings

Although randomized controlled studies have failed to demonstrate a beneficial effect of contact force (CF) guidance on AF-free survival at 1 year,^{14,15} further optimization of CF-guided ablation by adding objective criteria (the 'CLOSE' criteria of inter-lesion distance -ILD- and ablation-index -AI-) further improves acute and long-term isolation rates and 1-year single-procedure outcomes.¹⁶ 'CLOSE'-guided PVI improves procedural and 1-year outcomes in CF-guided PVI while shortening procedure time. These improvements cannot be explained by CF variability and are most likely due to the strict application of criteria for contiguity and RF lesion quality. Several observations testify to the reliability of the reported >90% single-procedure freedom from paroxysmal AF after 'CLOSE'-guided PVI: avoiding weak links within the RF circle deployed is the key to durable PV isolation and clinical success.^{17–20} Moreover, in a multi-center clinical practice setting, an ablation protocol that respected strict criteria for contiguity and the quality of lesions resulted in a high rate of 1-year freedom from AF recurrence and a low complication rate, irrespective of the ablation catheters used and the AI settings.^{21,22}

The currently reported high rate of single-procedure freedom from paroxysmal AF after PV isolation is in line with our results and is probably the best that one can expect after durable PV isolation.

4.3 | The role of LI in guiding successful ablation

LI measurements were found to well distinguish contact with healthy myocardium from contact with blood, and high-voltage areas from low-voltage areas; they were also able to detect significant differences in LI drop during ablation according to the voltage level. This approach may therefore provide useful indications for the creation of

effective lesions.^{10–12} In our study, successful ablation sites were predicted by both baseline LI and LI drop during RF ablation, a value greater than 14 Ω being associated with effective sites, in agreement with the literature.^{10–12} In two studies,^{12,23} LI was reported to be a sensitive parameter for guiding radiofrequency ablation in patients with ventricular tachycardia (VT) when VT termination or loss of capture in the left ventricle was considered as the ablation endpoint. These two studies yielded comparative values and provided an indicative target range of 16 to 18.5 Ω for LI: successful ablation was associated with a drop rate of 0.65 Ω/s (which the authors indicated as an initial reference value for LI-guided ablation). In the present study, on the basis of the ROC analysis, the cut-off value of LI drop/ τ that best predicted successful RF delivery was 0.65 Ω/s ; a LI drop/ τ greater than this value was significantly associated with successful RF delivery, with an OR of 5.54.

In the LOCALIZE trial, blood pool and pre-ablation LI were predictive of acute lesion formation, as measured by LI drop during PVI in 60 patients with paroxysmal AF. A 3-month remapping procedure in 44 patients showed that segments with ILD ≤ 6 mm and a ≥ 16.6 Ω LI drop were associated with durable PV block. In addition, the regional analysis showed that achieving an LI drop of ≥ 16.9 Ω in anterior/superior segments and an LI drop of ≥ 14.2 Ω in posterior/inferior segments was highly predictive of durable segment block.²⁴ In agreement with these findings, our analysis showed that a value greater than 15 Ω was consistently associated with acute success.

When considering the optimal decrease in LI, safety aspects need to be assessed. Both a more rapid initial Generator Impedance reduction during catheter ablation (1.40 Ω/s with steam pops vs. 0.38 Ω/s without)²⁵ and a larger Δ LI during ablation (the smallest Δ LI that was associated with a steam pop was 37 Ω) correlated with steam pops.¹²

In our study, no steam pops or major complications, including atrioesophageal fistula or tamponade, occurred during or after the procedures. On the basis of our results and the above considerations, we think that an LI drop of 30 Ω might be proposed as a reasonable maximum cut-off value. About half of this value, 15 Ω , as reported by Das et al.²⁴ and in agreement with our findings, can be used as a predictor of a probably effective lesion. When used in conjunction with the local signal, LI could be a real-time measure of electrical coupling, allowing lesion size to be estimated during ablation, even when lower ILD and higher power are used.

5 | LIMITATIONS

The results of the present study should be interpreted in light of several limitations. First, this was not a randomized controlled study and, therefore, direct comparisons with other technologies cannot be made. Although a catheter providing a simultaneous combination of LI and contact-force measurements was under development, it was not available when the study was performed. As contact-force sensing was not available, we could not compare contact-force sensing with impedance sensing. A manual tagging of RF lesions was used

because the auto-tagging feature was not available at the time of study conduction. However, a graphical tag dimension equal to 6 mm was set, so the absence of continuity solutions was indicative of an inter-lesion <6 mm. We did not make an invasive analysis of all patients to look for reconnection, and ILDs were not reevaluated during our postprocessing analysis.

Therefore, we cannot provide data on the relationship between LI value and ILD with long-term PV reconnection. The LI values that we used were empirically chosen. However, they were based on our previous experience in clinical practice, in which they had allowed us to achieve considerable clinical success. The reduction in signal voltage by at least 50% and the inability to capture local tissue on pacing may not fully represent the concept of effective RF ablation. However, several works in this field refer to this definition and several authors applied the proposed approach. We analyzed 3556/7598 point-by-point first-pass RF applications of >10 s duration because a complete video recording was available only in 105 (69%) cases. Despite this limitation, at the present time, this is the largest analysis of LI-guided ablation lesions. For the reason of excluding RF deliveries outside PVs from the analysis, we cannot provide differences in impedance drop between anterior/posterior wall, roof, ridge

6 | CONCLUSIONS

In our study, the magnitude and the time-course of the LI drop during RF delivery were associated with effective lesion formation. This ablation strategy for PVI guided by LI technology proved safe and effective and resulted in a very low rate of AF recurrence over 1-year follow-up.

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CONFLICT OF INTERESTS

M.M. is an employee of Boston Scientific. For the remaining authors, there are no conflict of interests.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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